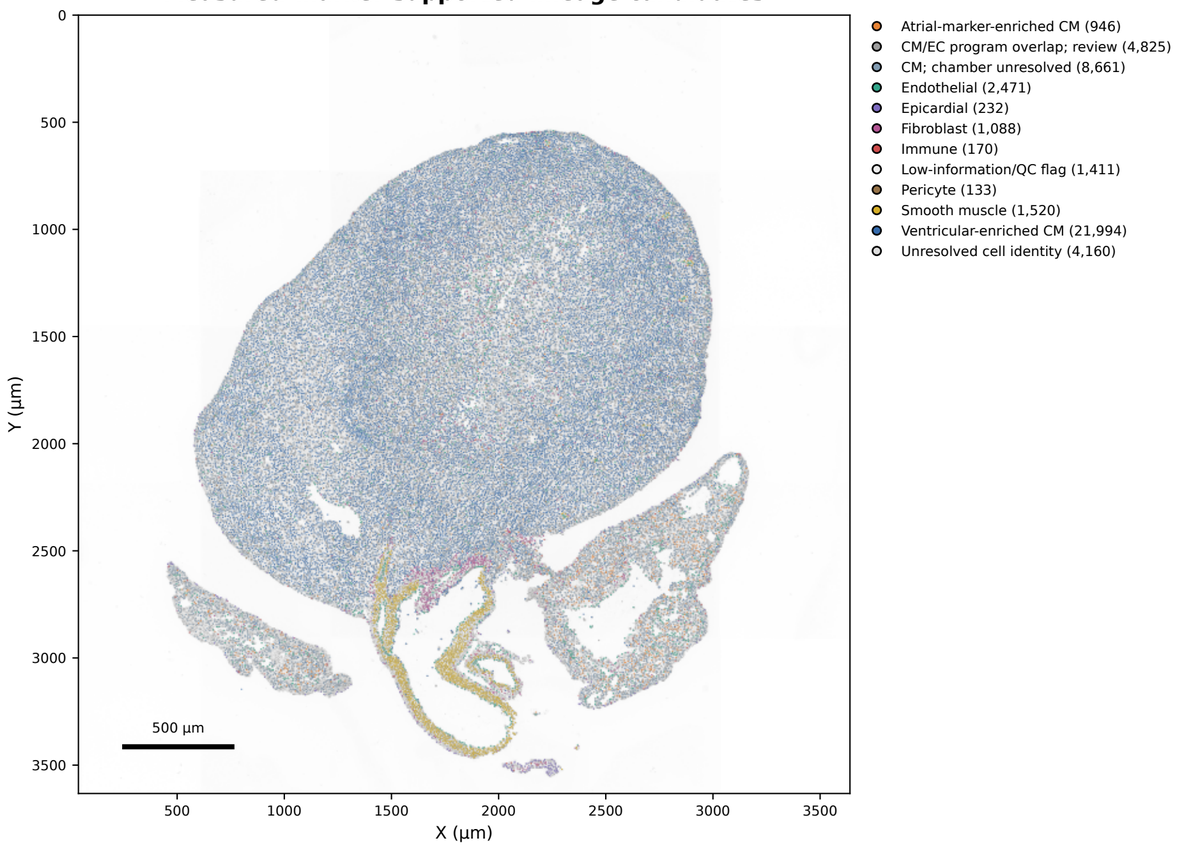
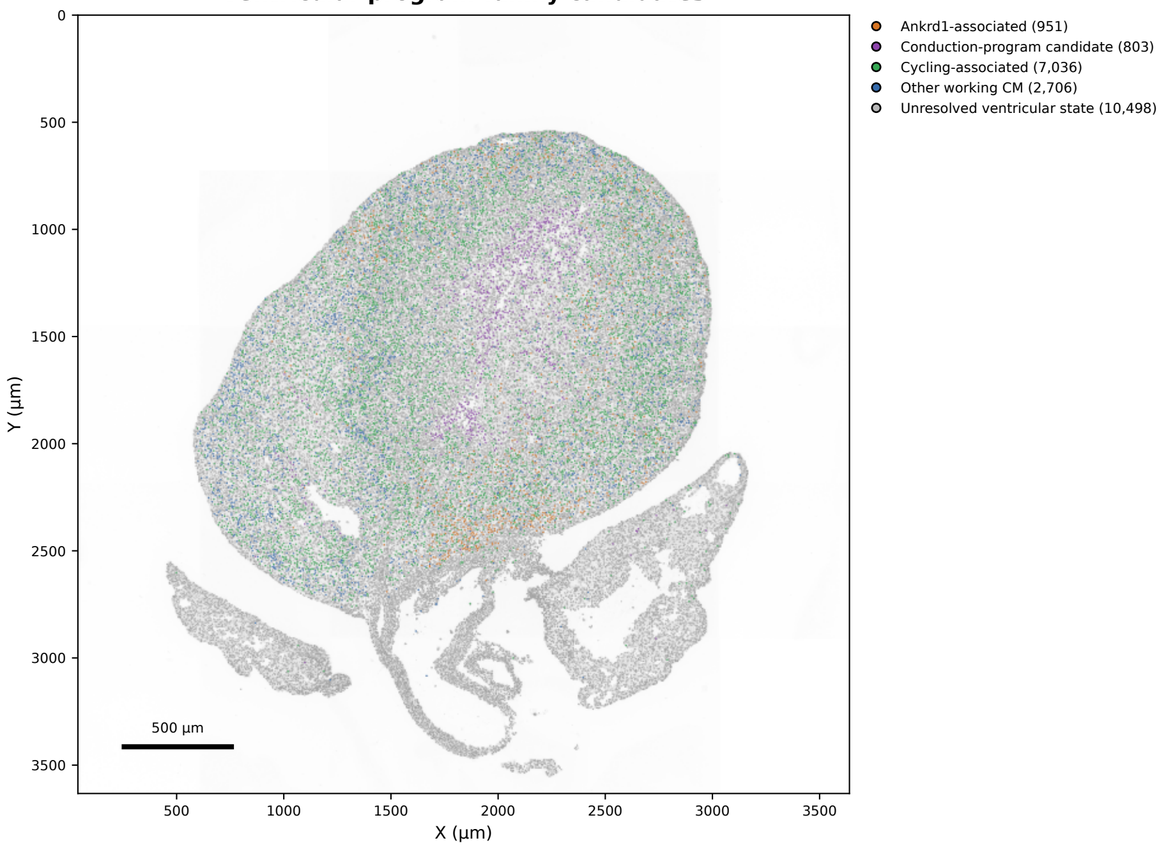
P0: ventricular candidates on the acquired cardiac section

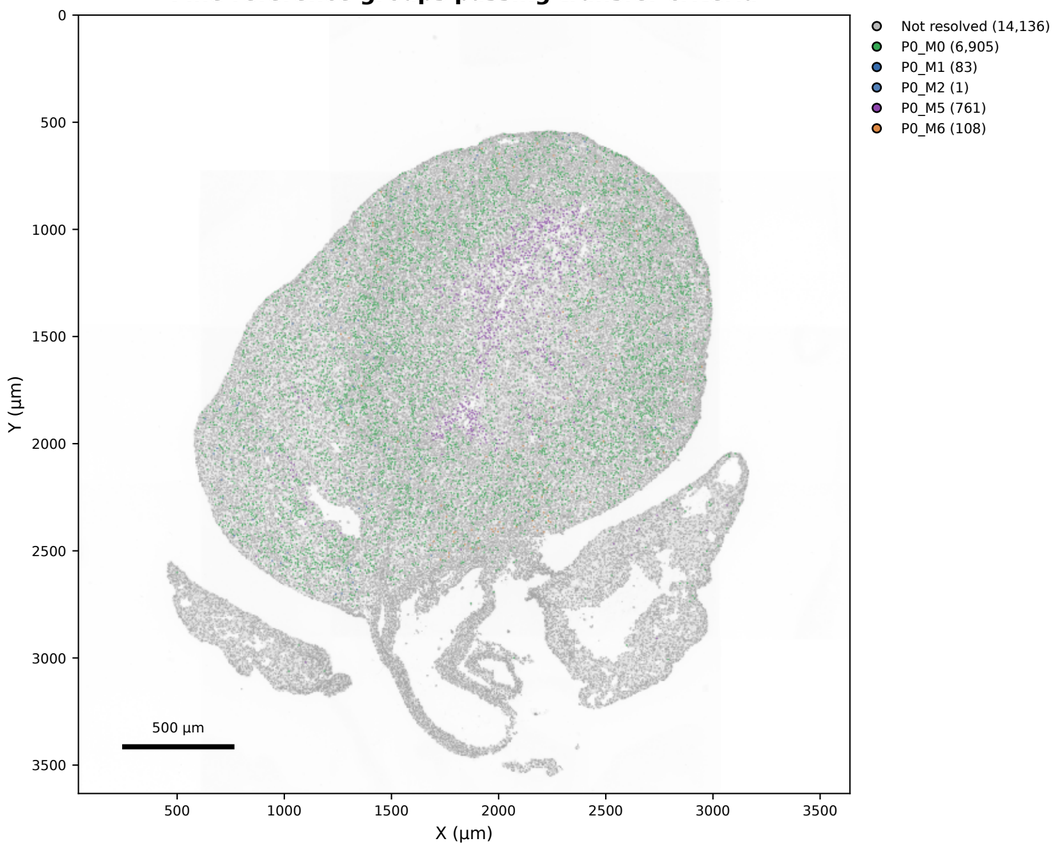
Measured-marker-supported lineage candidates



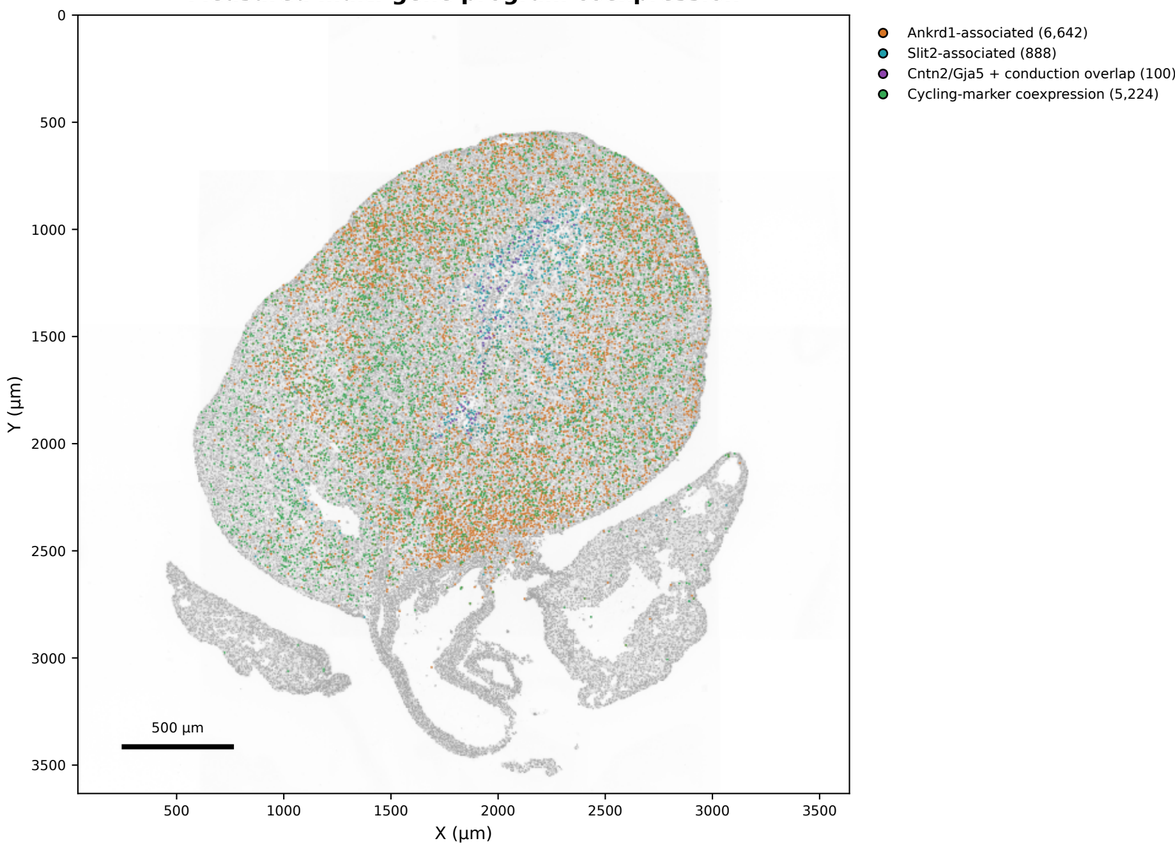
Ventricular program-family candidates



Fine reference groups passing transfer criteria

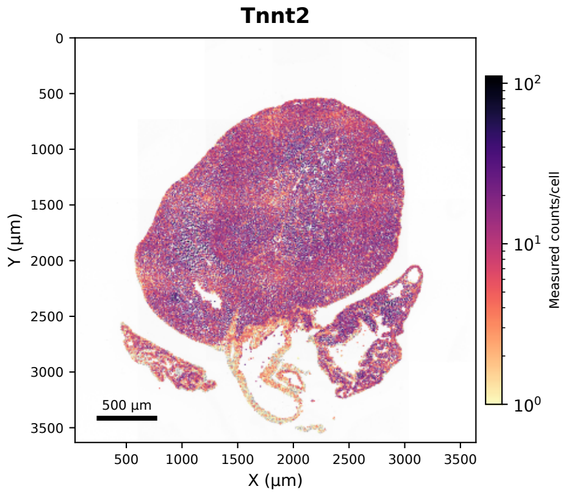


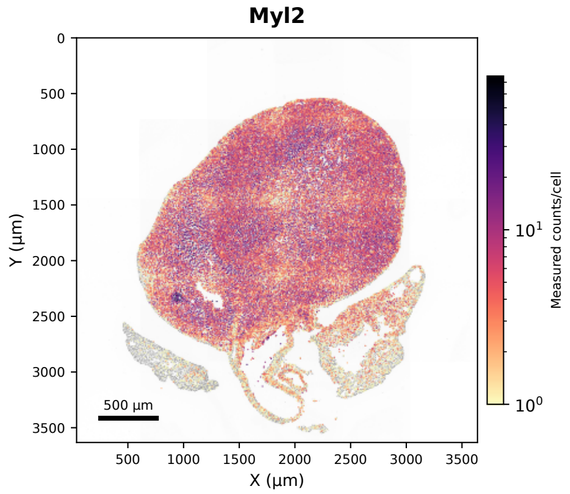
Measured multi-gene program coexpression

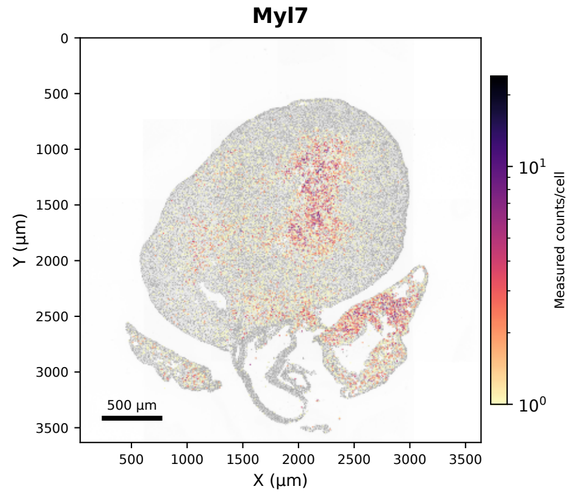


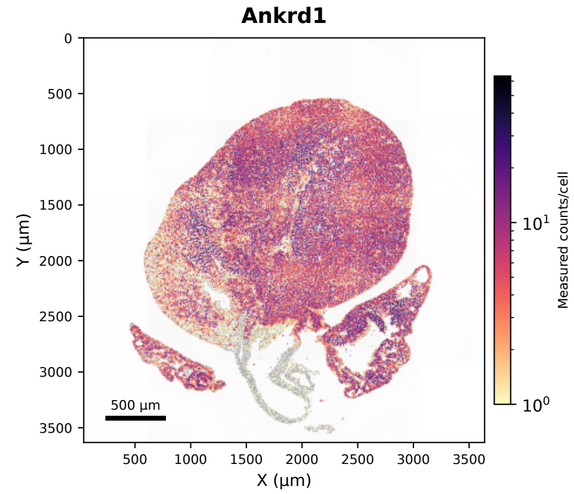
Gray image: acquired DAPI; colored points: original cell centroids. Candidate labels are reference-assisted, not anatomical or functional verification.
Fine assignments require panel/library/raw-reference agreement. Unresolved cells remain gray; measured program flags can overlap.
Kcnd2 and Irx5 are NOT assayed in this Xenium panel. No Kcnd2 expression, imputation or transmural coordinate is shown.

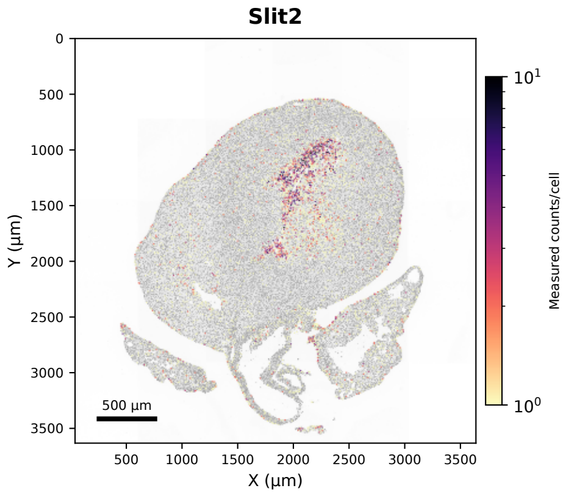
P0: directly measured Xenium RNA on DAPI

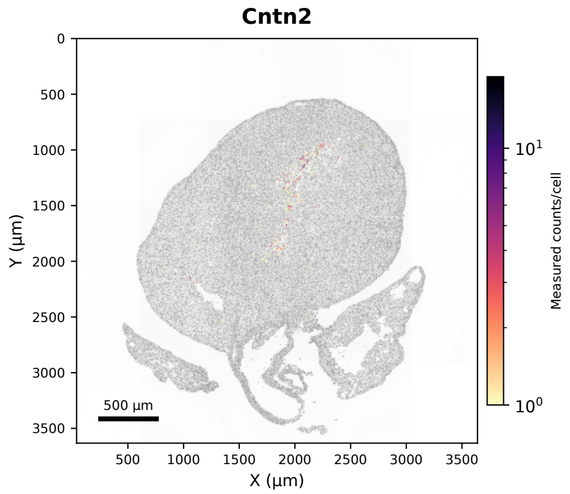


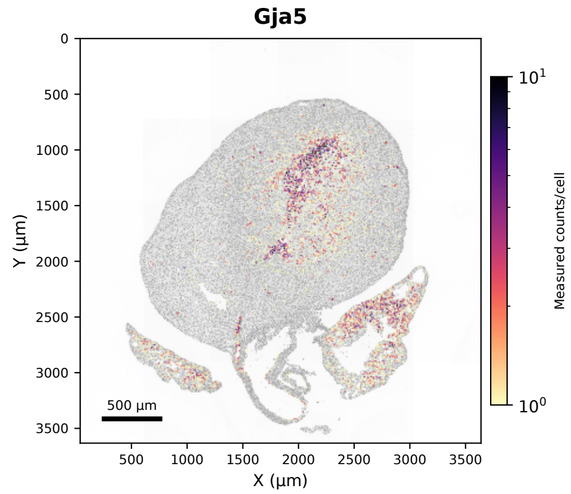


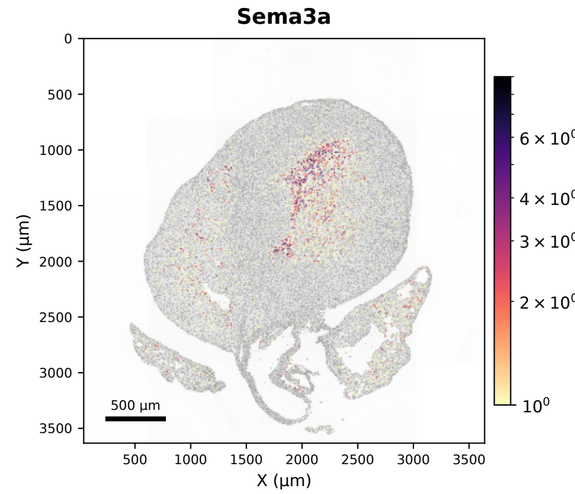


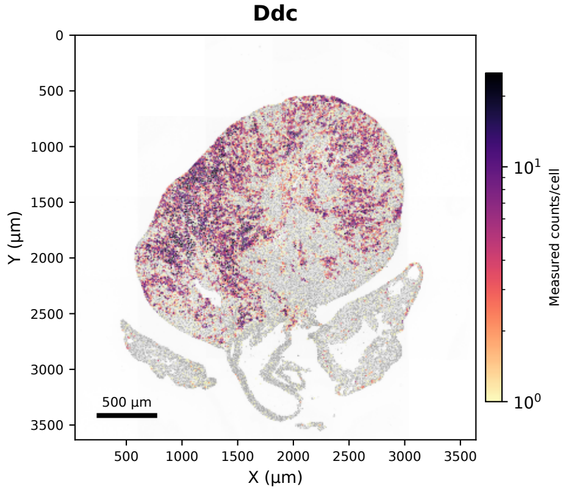


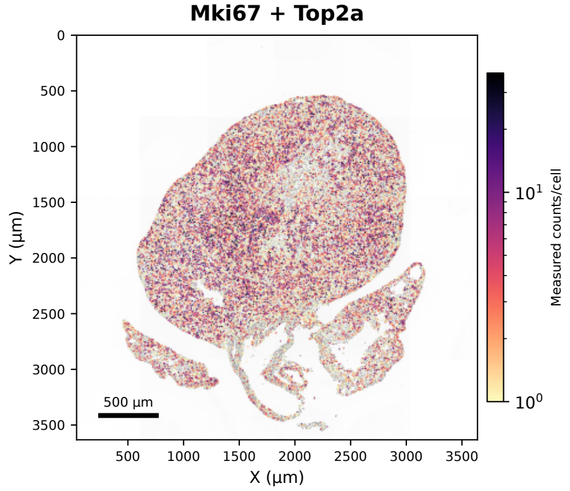


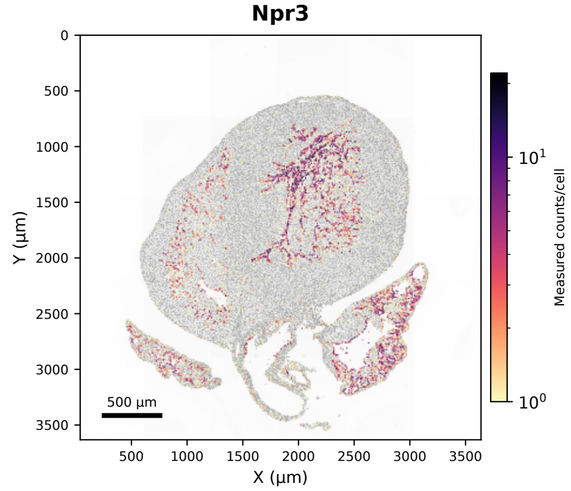


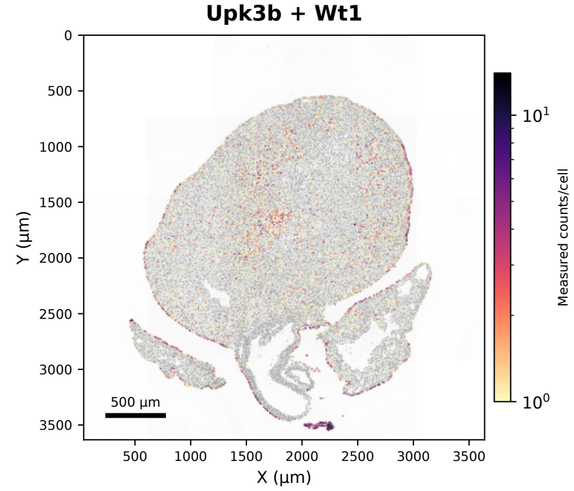






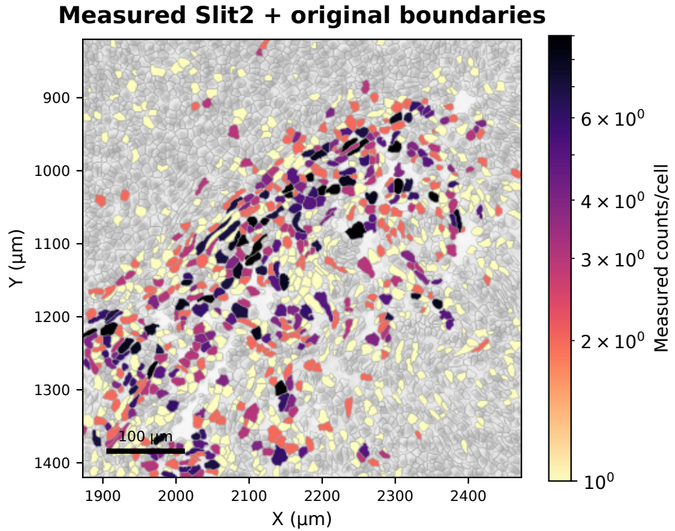


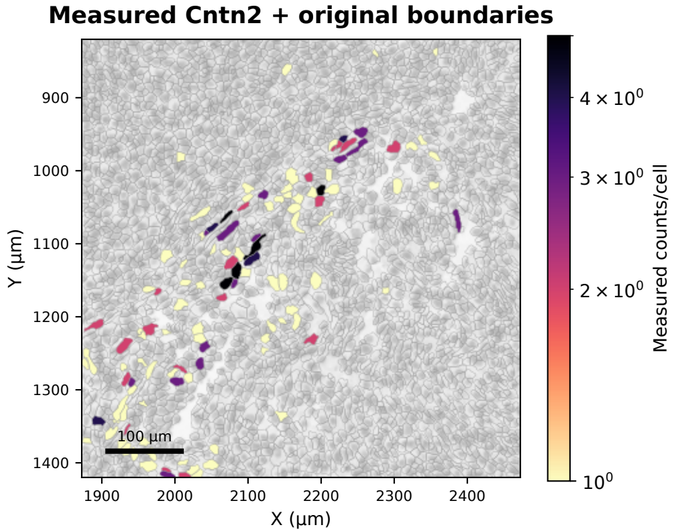


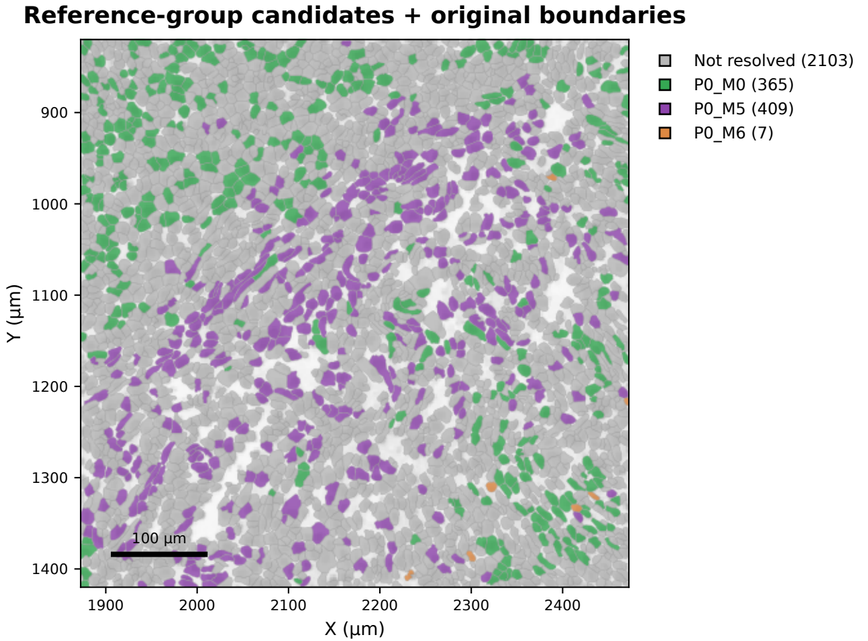


All QC-pass segmented cells shown; these genes are not all CM-specific (notably Gja5). Same gene uses the same scale across stages.
No count correction, smoothing or expression imputation. Marker counts depend on cell area, segmentation and detection efficiency.
Npr3 and Upk3b/Wt1 offer candidate anatomical landmarks; they are not traced wall boundaries. Kcnd2/Irx5 are absent from the panel.

P0: 600 × 600 µm illustrative region



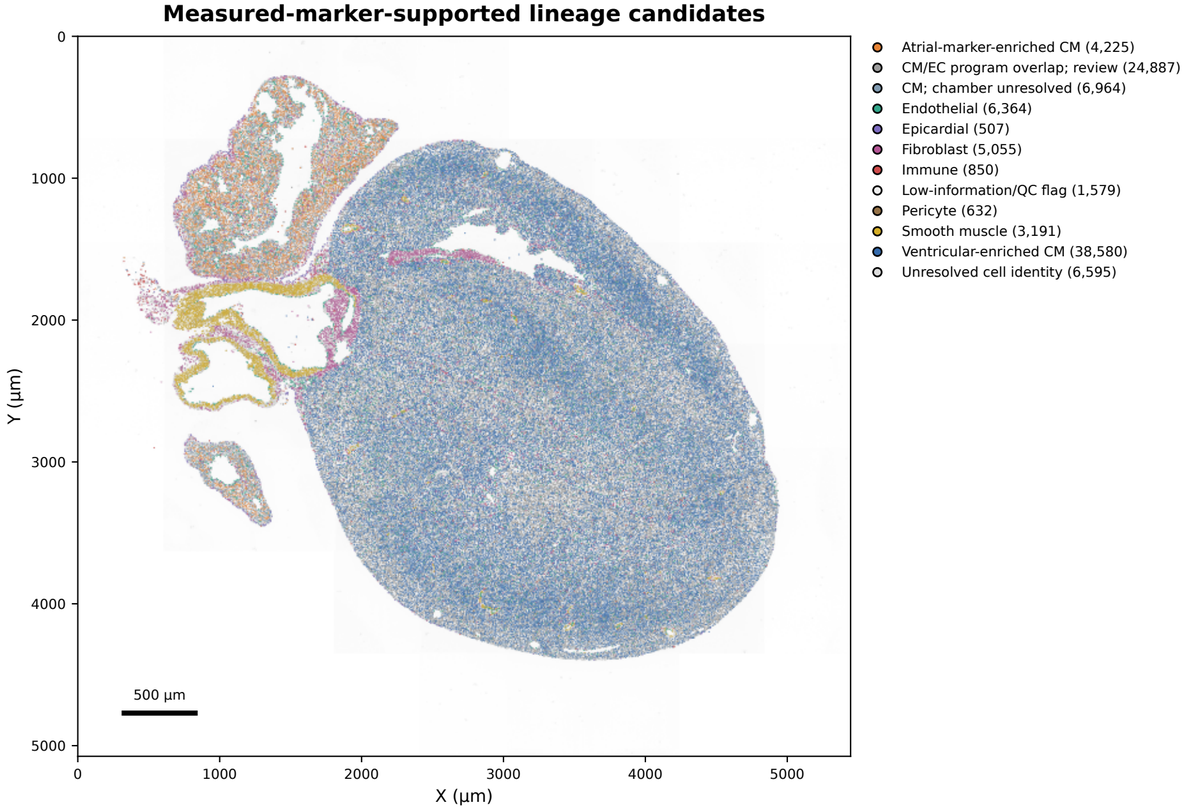




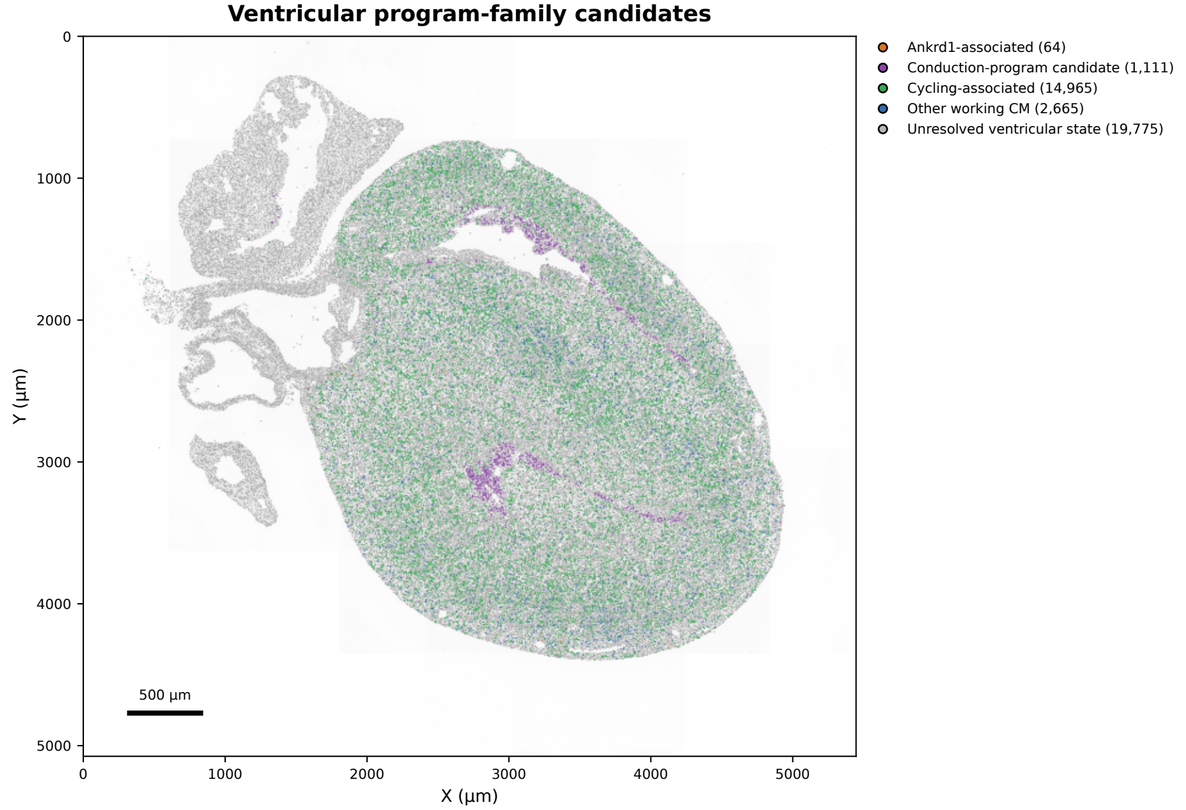
ROI selected by density of reference-assisted conduction-program candidates; not an unbiased enrichment test.
Original boundaries and RNA counts retained; labels do not verify Purkinje identity. Kcnd2 is not measured.

P7: ventricular candidates on the acquired cardiac section

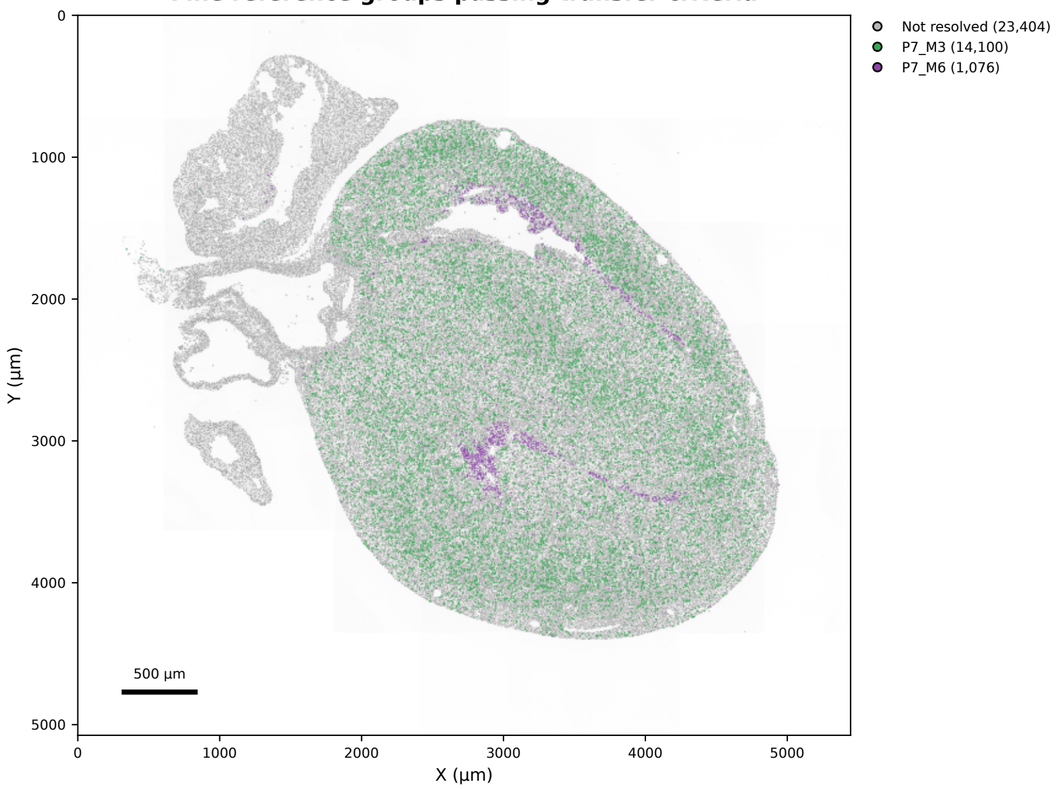
Measured-marker-supported lineage candidates



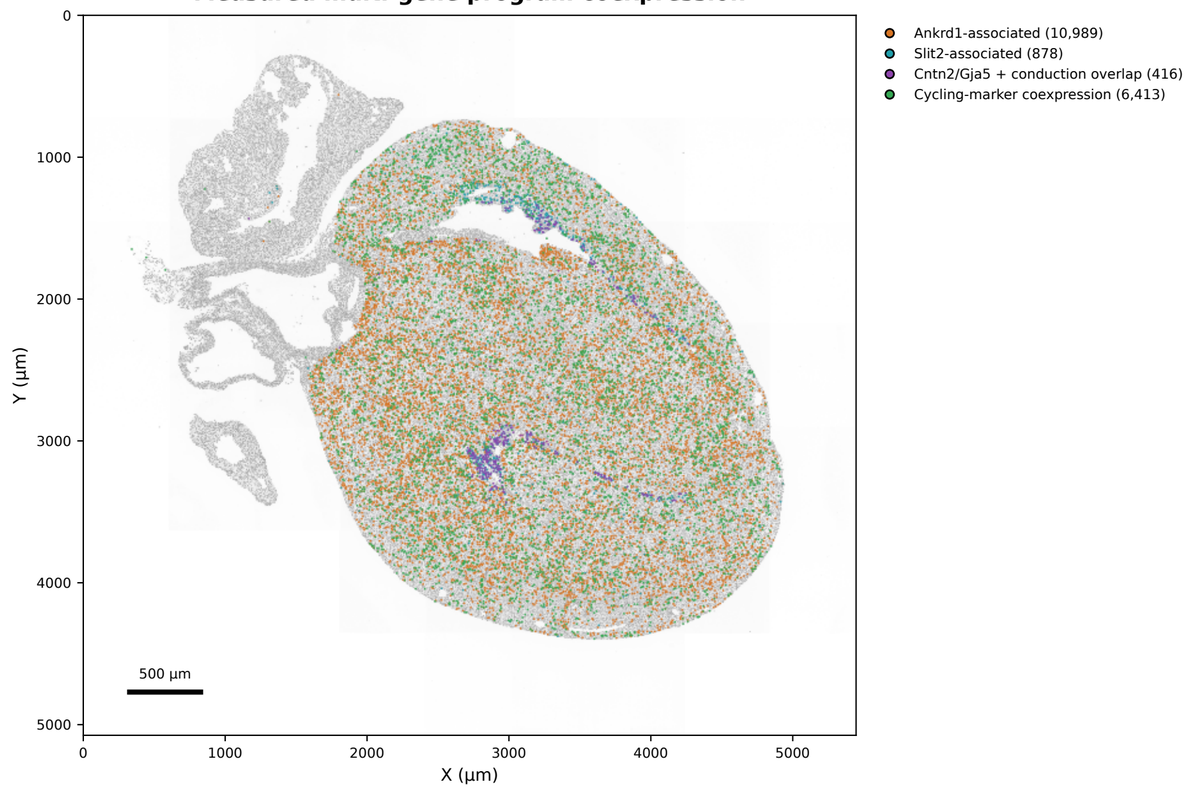
Ventricular program-family candidates



Fine reference groups passing transfer criteria

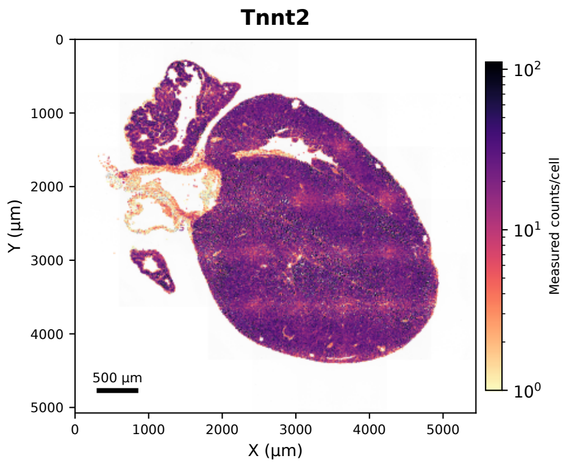


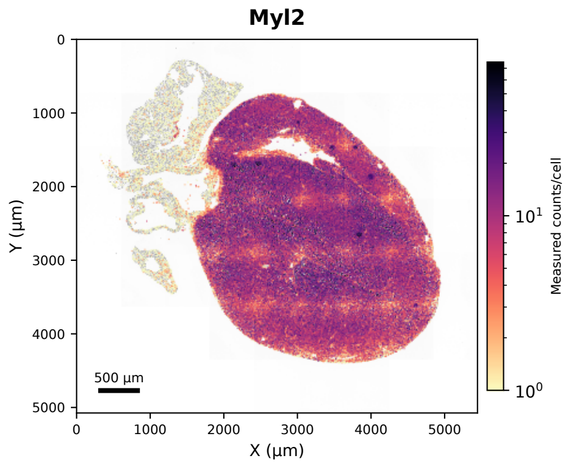
Measured multi-gene program coexpression

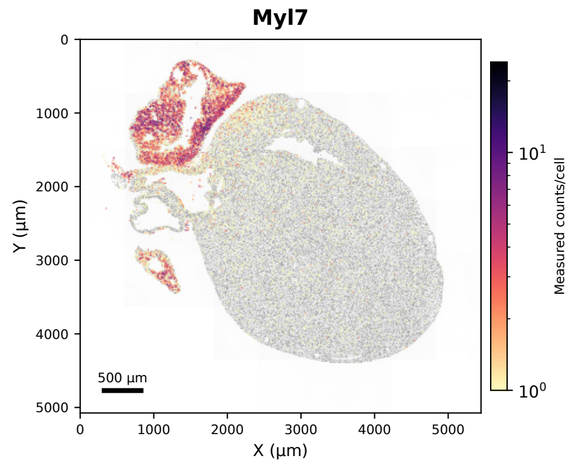


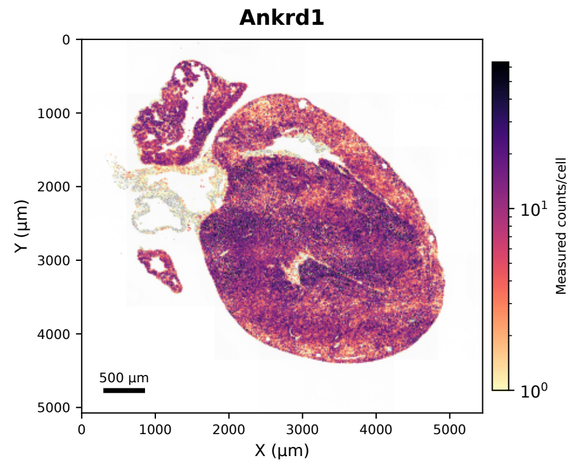
Gray image: acquired DAPI; colored points: original cell centroids. Candidate labels are reference-assisted, not anatomical or functional verification.
Fine assignments require panel/library/raw-reference agreement. Unresolved cells remain gray; measured program flags can overlap.
Kcnd2 and Irx5 are NOT assayed in this Xenium panel. No Kcnd2 expression, imputation or transmural coordinate is shown.

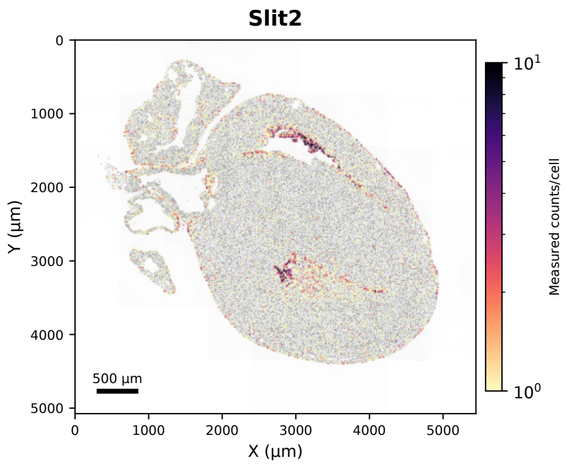
P7: directly measured Xenium RNA on DAPI

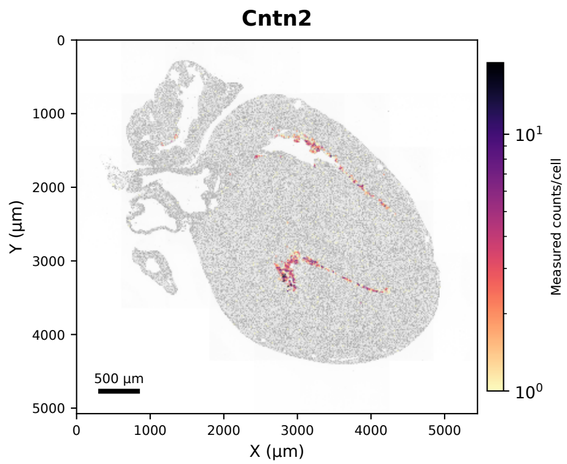


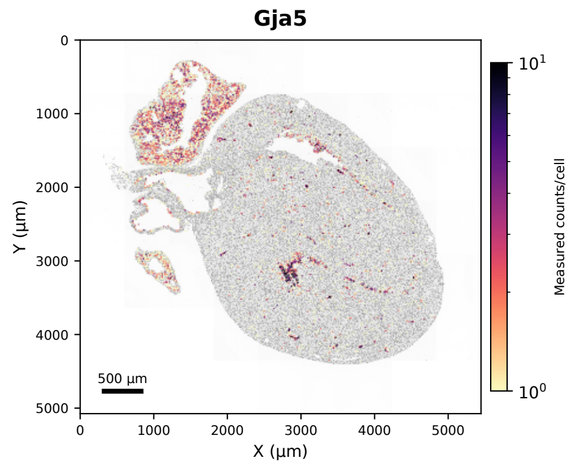


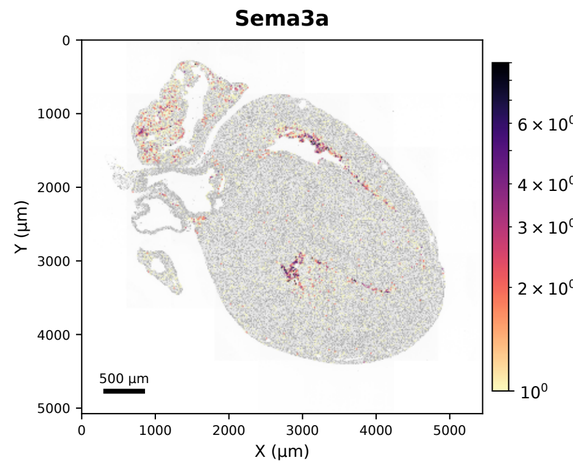


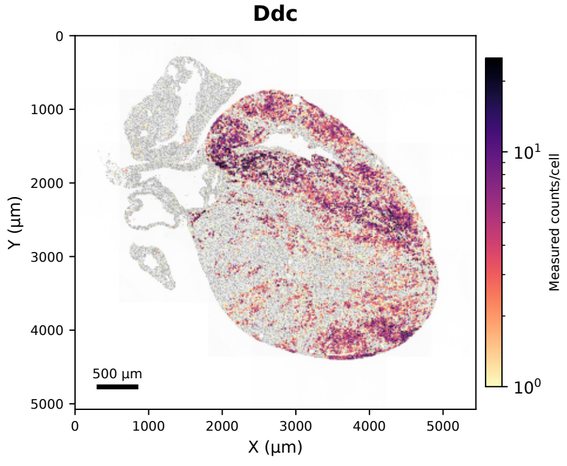


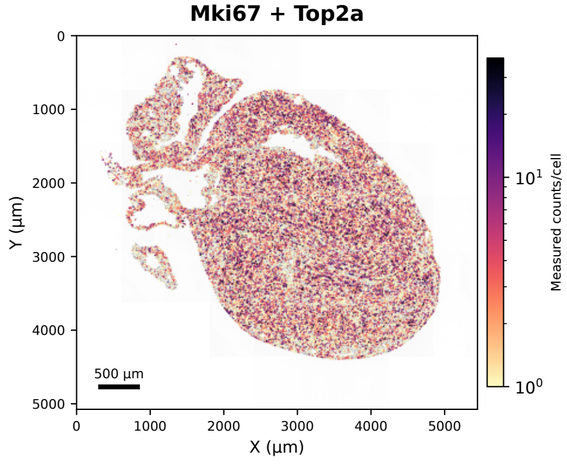


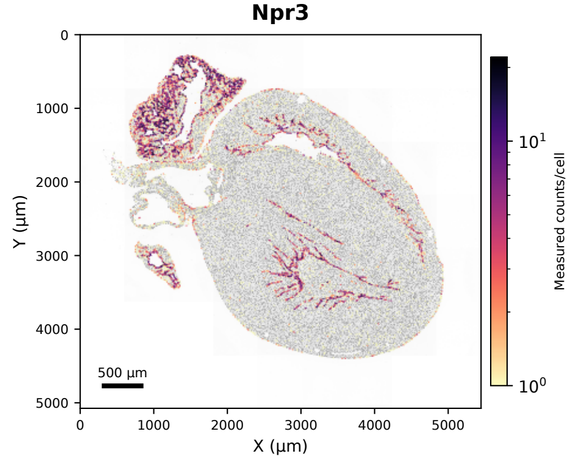


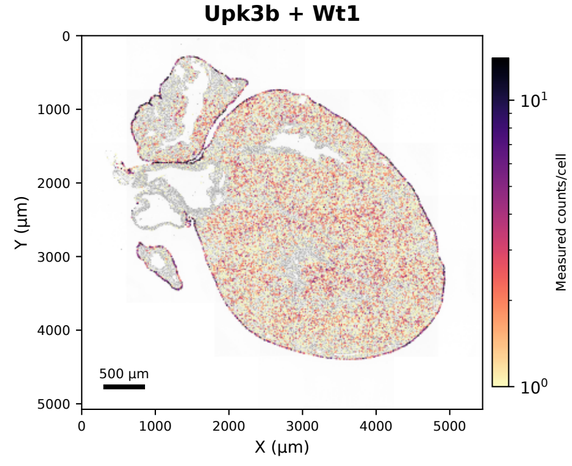






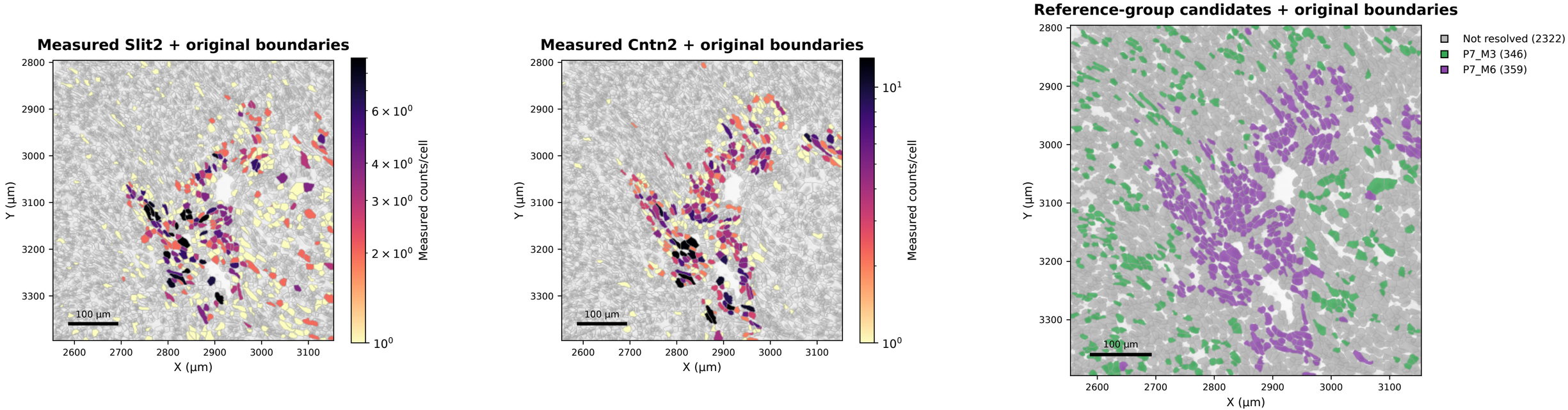






All QC-pass segmented cells shown; these genes are not all CM-specific (notably Gja5). Same gene uses the same scale across stages.
No count correction, smoothing or expression imputation. Marker counts depend on cell area, segmentation and detection efficiency.
Npr3 and Upk3b/Wt1 offer candidate anatomical landmarks; they are not traced wall boundaries. Kcnd2/Irx5 are absent from the panel.

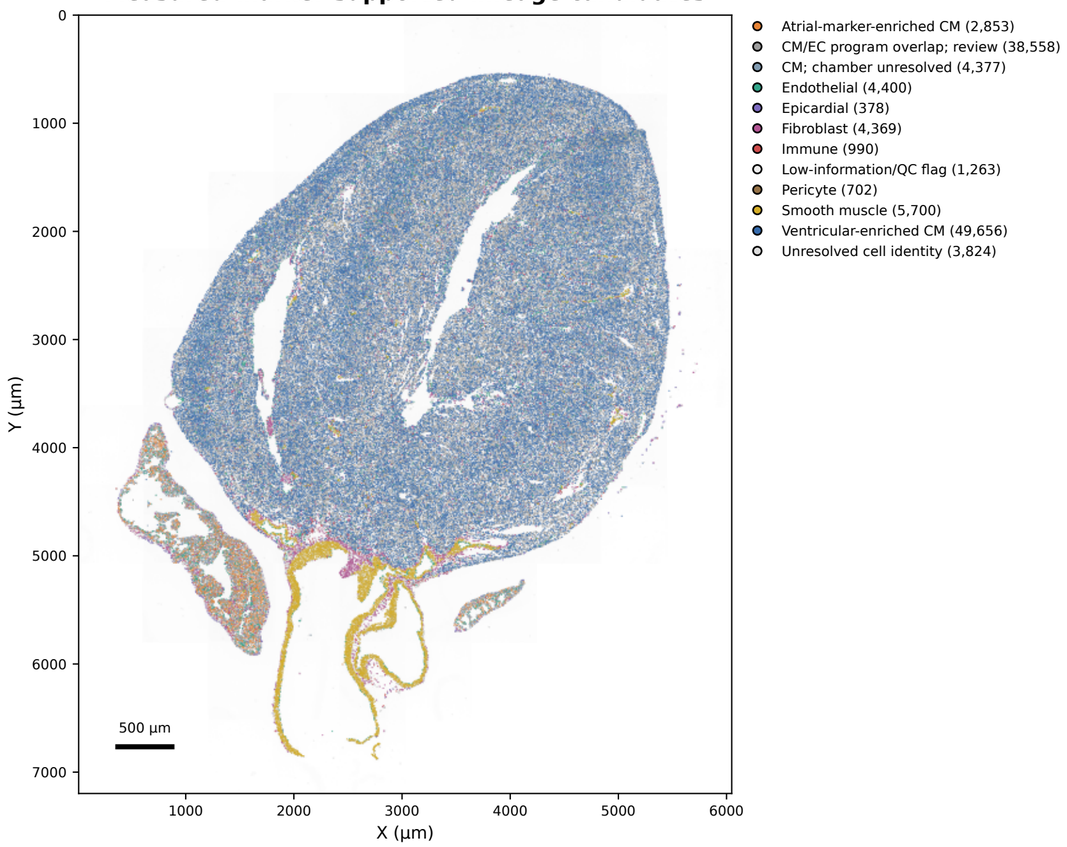
P7: 600 × 600 µm illustrative region



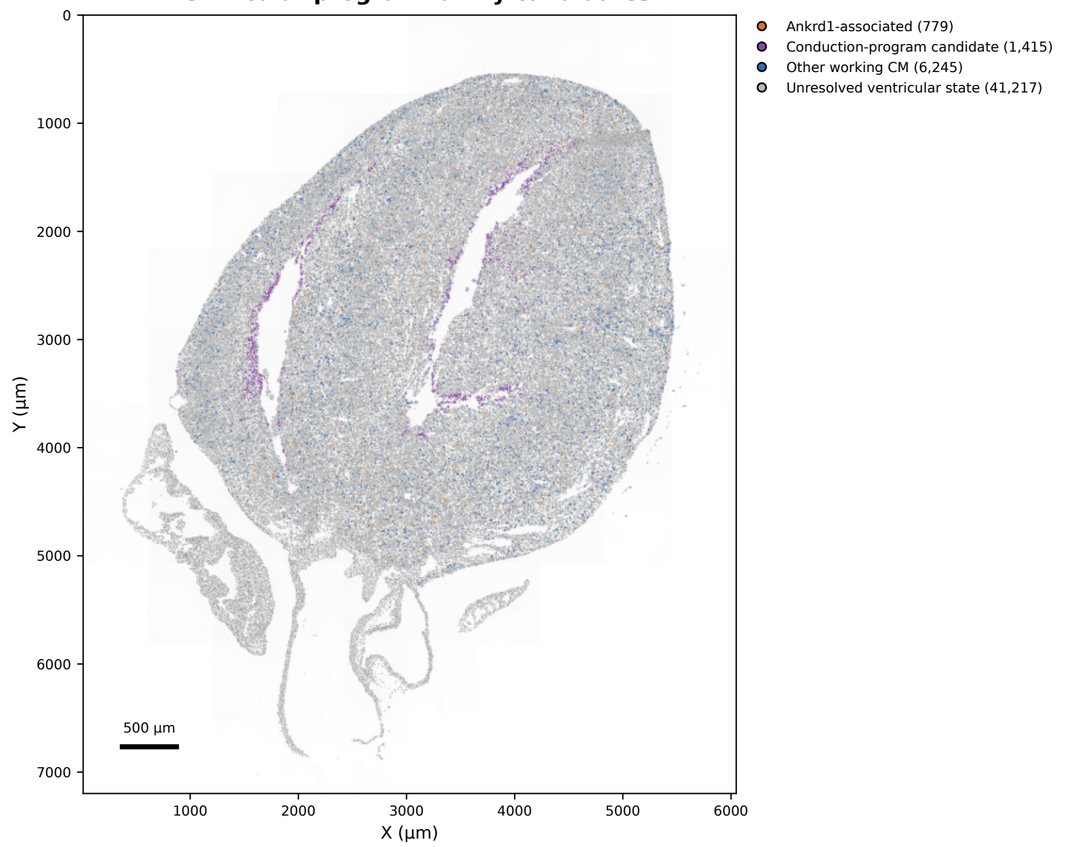
ROI selected by density of reference-assisted conduction-program candidates; not an unbiased enrichment test.
Original boundaries and RNA counts retained; labels do not verify Purkinje identity. Kcnd2 is not measured.

P14: ventricular candidates on the acquired cardiac section

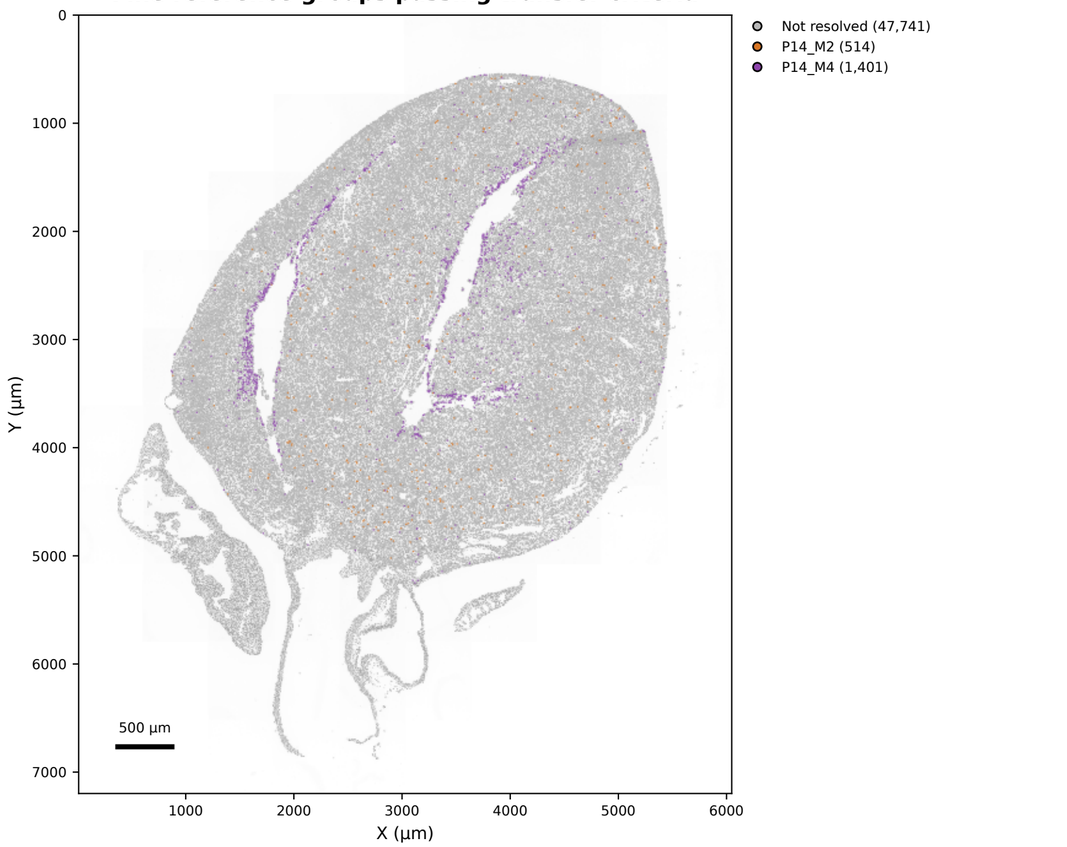
Measured-marker-supported lineage candidates



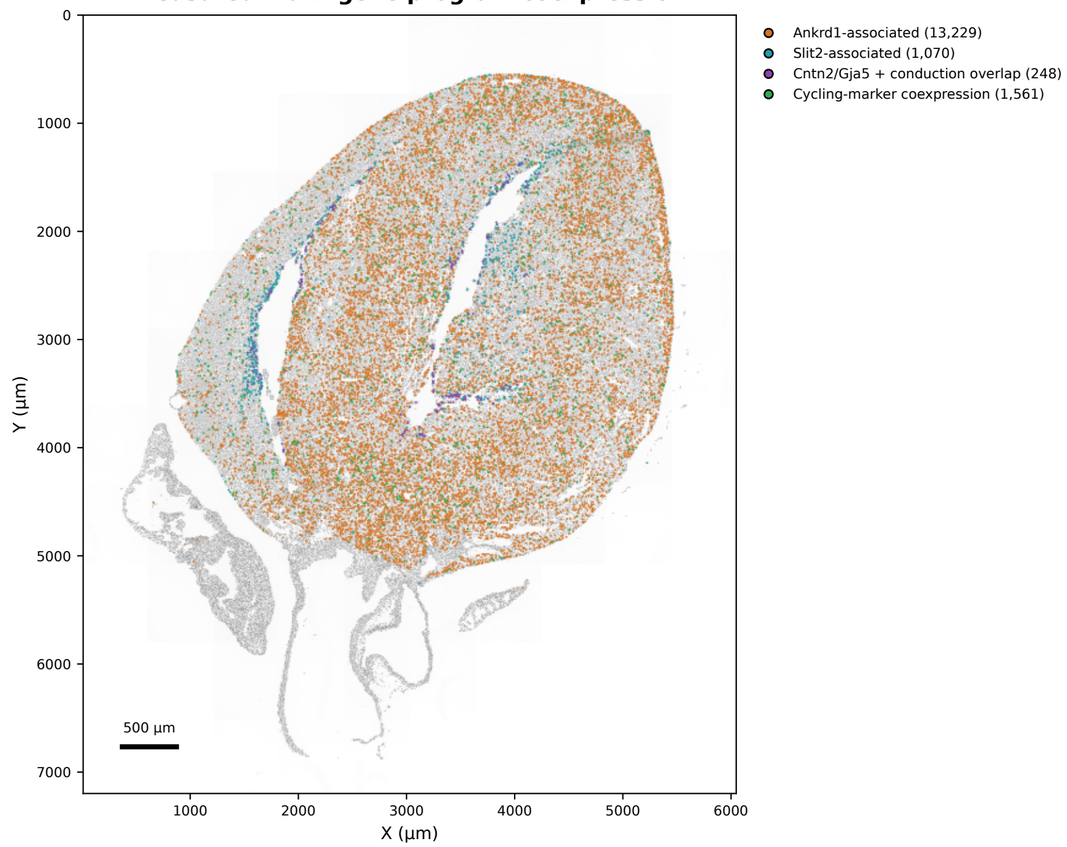
Ventricular program-family candidates



Fine reference groups passing transfer criteria

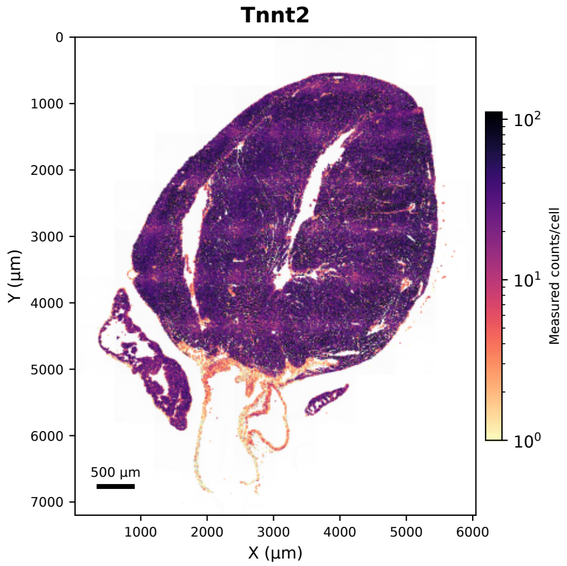


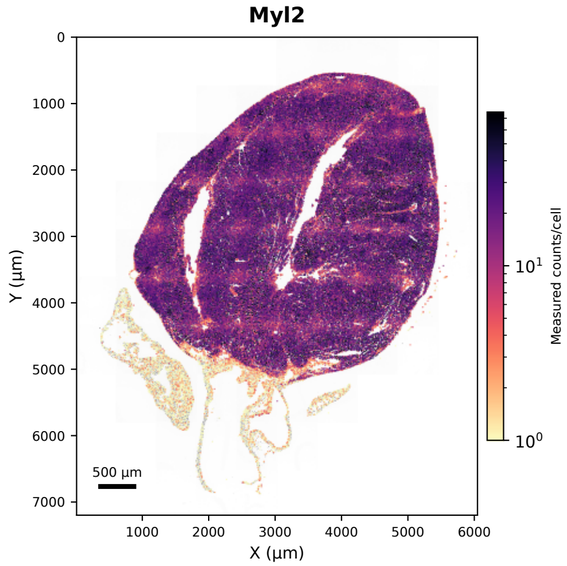
Measured multi-gene program coexpression

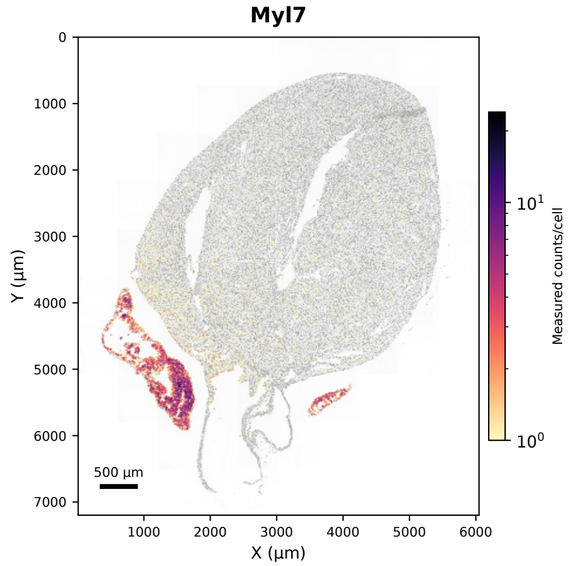


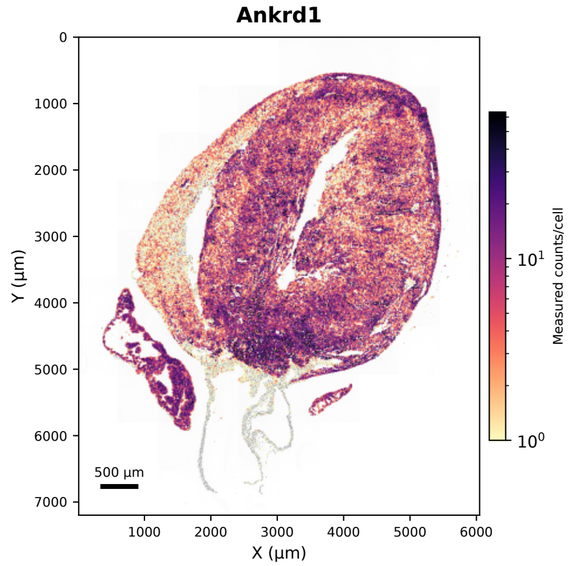
Gray image: acquired DAPI; colored points: original cell centroids. Candidate labels are reference-assisted, not anatomical or functional verification.
Fine assignments require panel/library/raw-reference agreement. Unresolved cells remain gray; measured program flags can overlap.
Kcnd2 and Irx5 are NOT assayed in this Xenium panel. No Kcnd2 expression, imputation or transmural coordinate is shown.

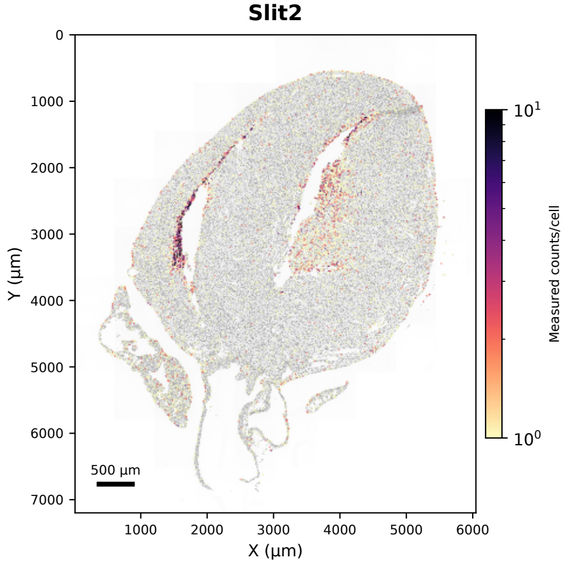
P14: directly measured Xenium RNA on DAPI

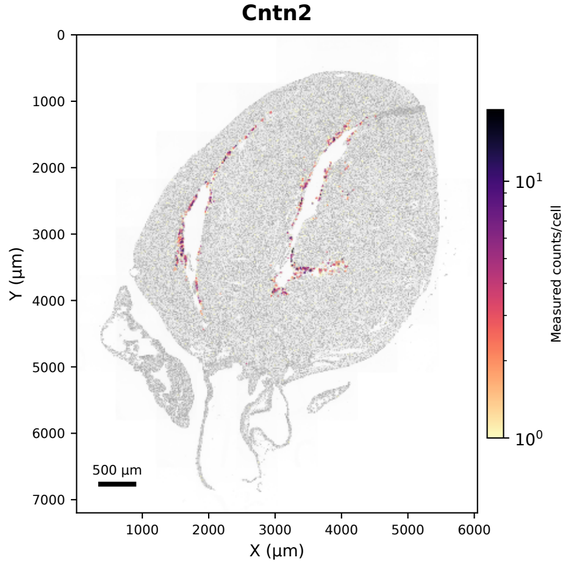


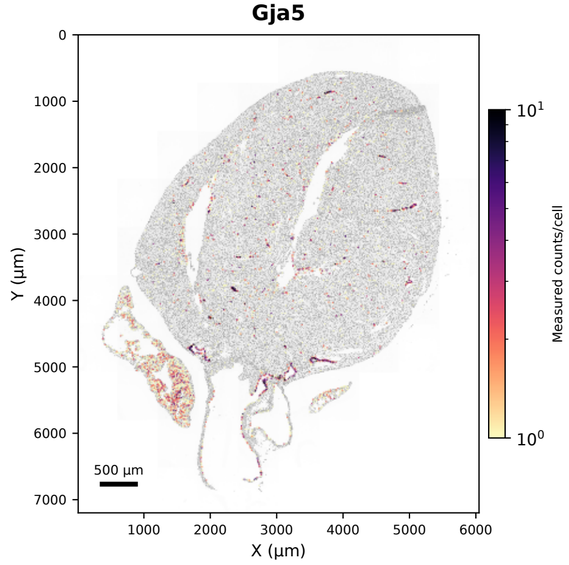


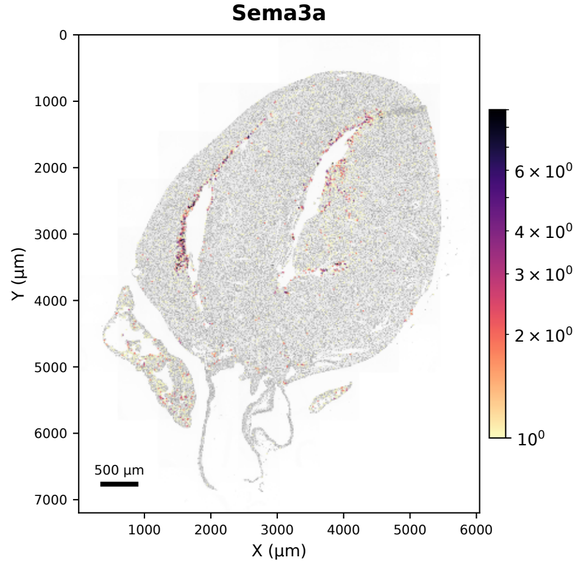


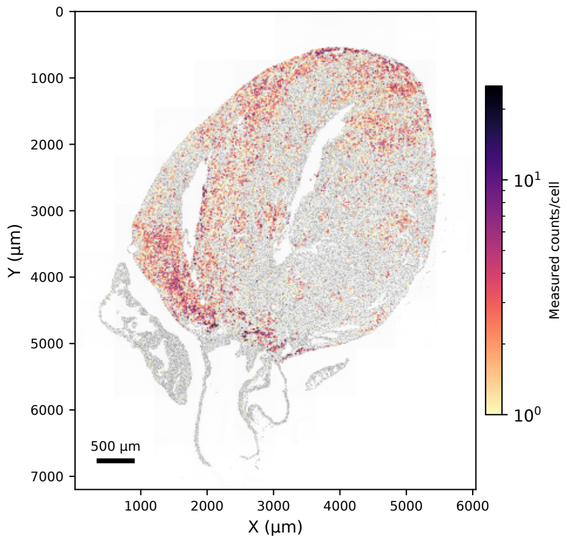


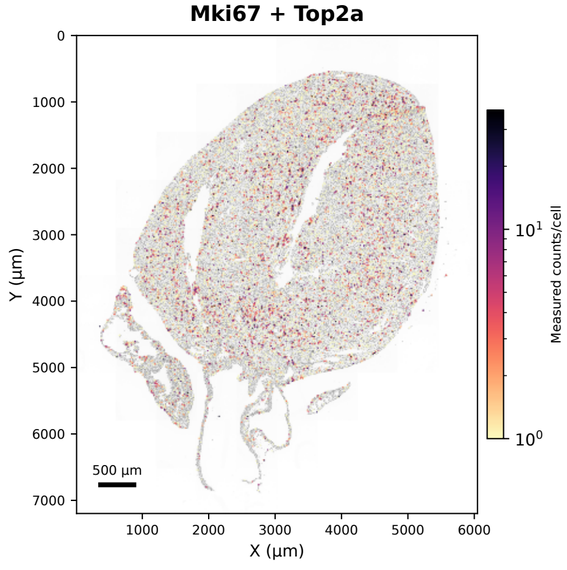


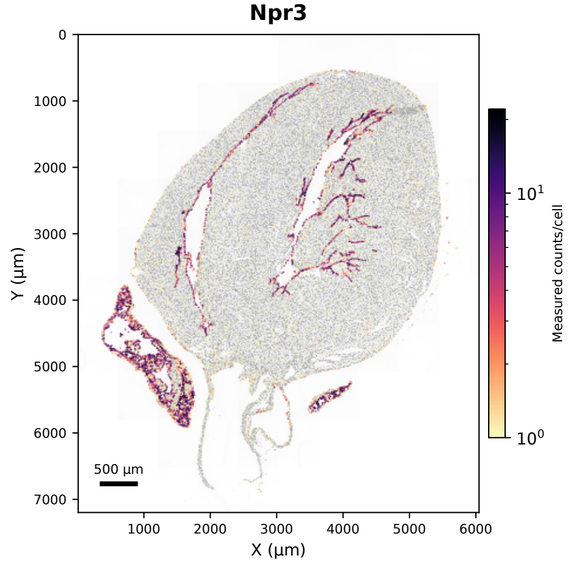


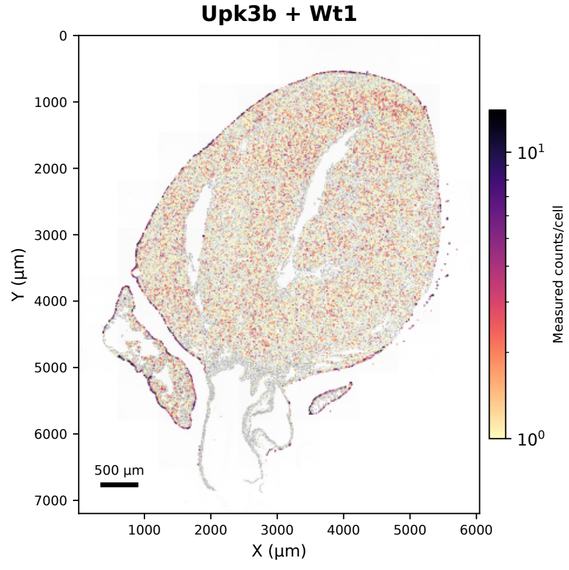






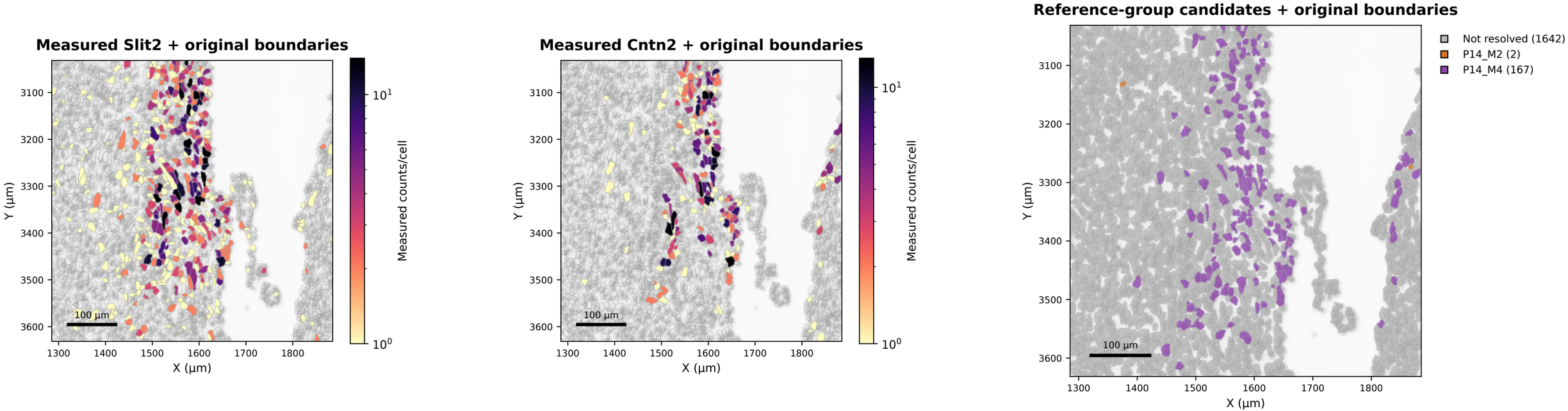






All QC-pass segmented cells shown; these genes are not all CM-specific (notably Gja5). Same gene uses the same scale across stages.
No count correction, smoothing or expression imputation. Marker counts depend on cell area, segmentation and detection efficiency.
Npr3 and Upk3b/Wt1 offer candidate anatomical landmarks; they are not traced wall boundaries. Kcnd2/Irx5 are absent from the panel.

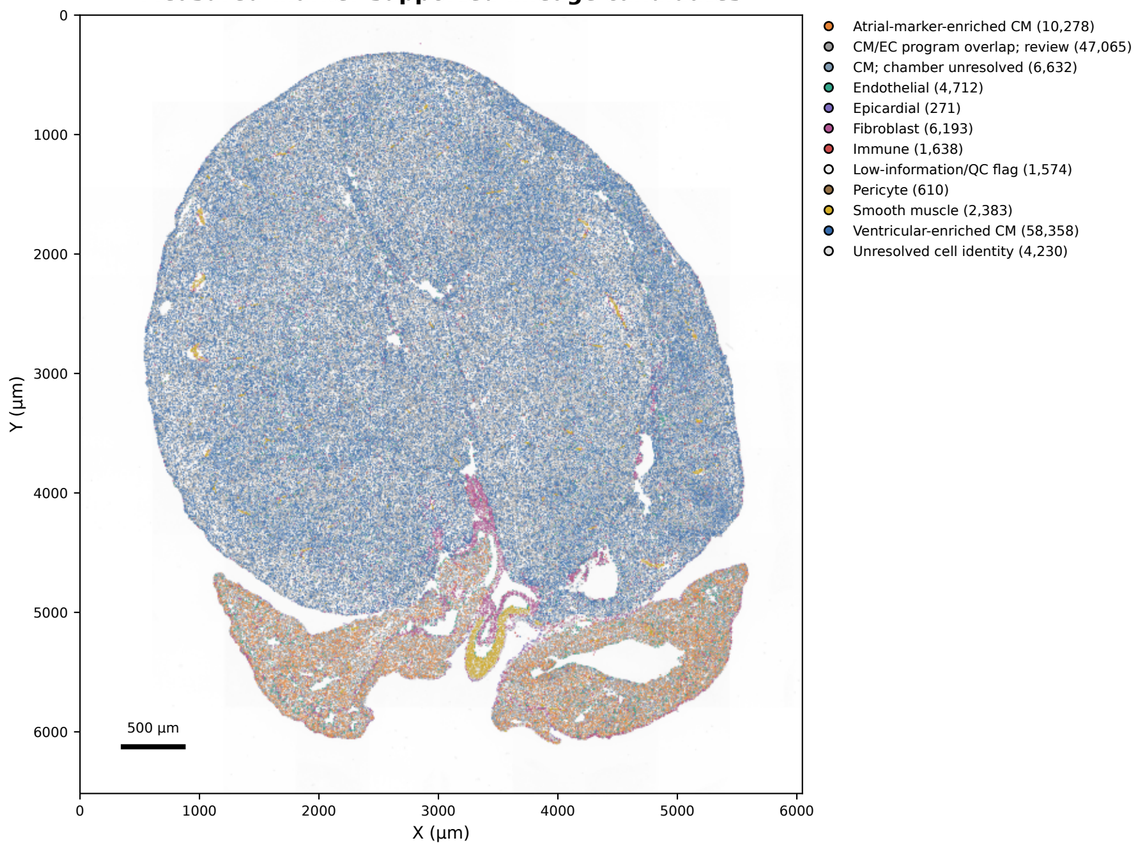
P14: 600 × 600 µm illustrative region



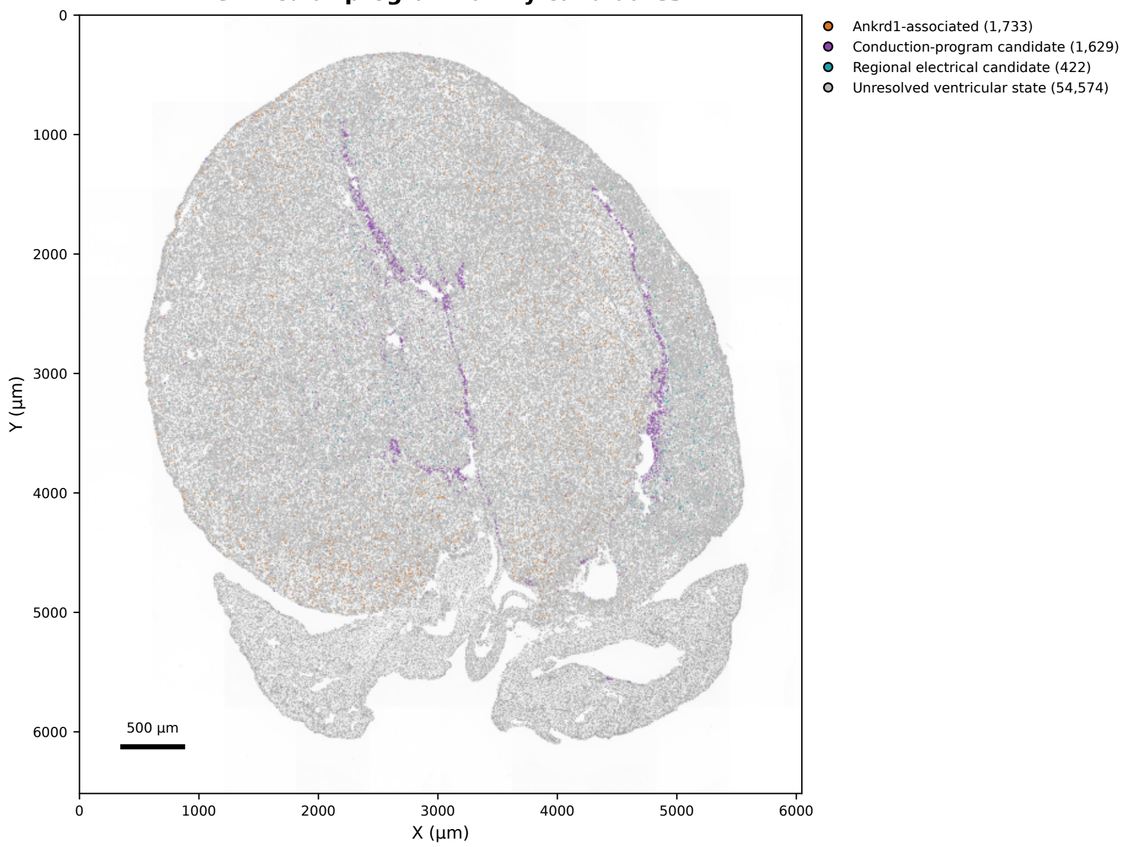
ROI selected by density of reference-assisted conduction-program candidates; not an unbiased enrichment test.
Original boundaries and RNA counts retained; labels do not verify Purkinje identity. Kcnd2 is not measured.

P21: ventricular candidates on the acquired cardiac section

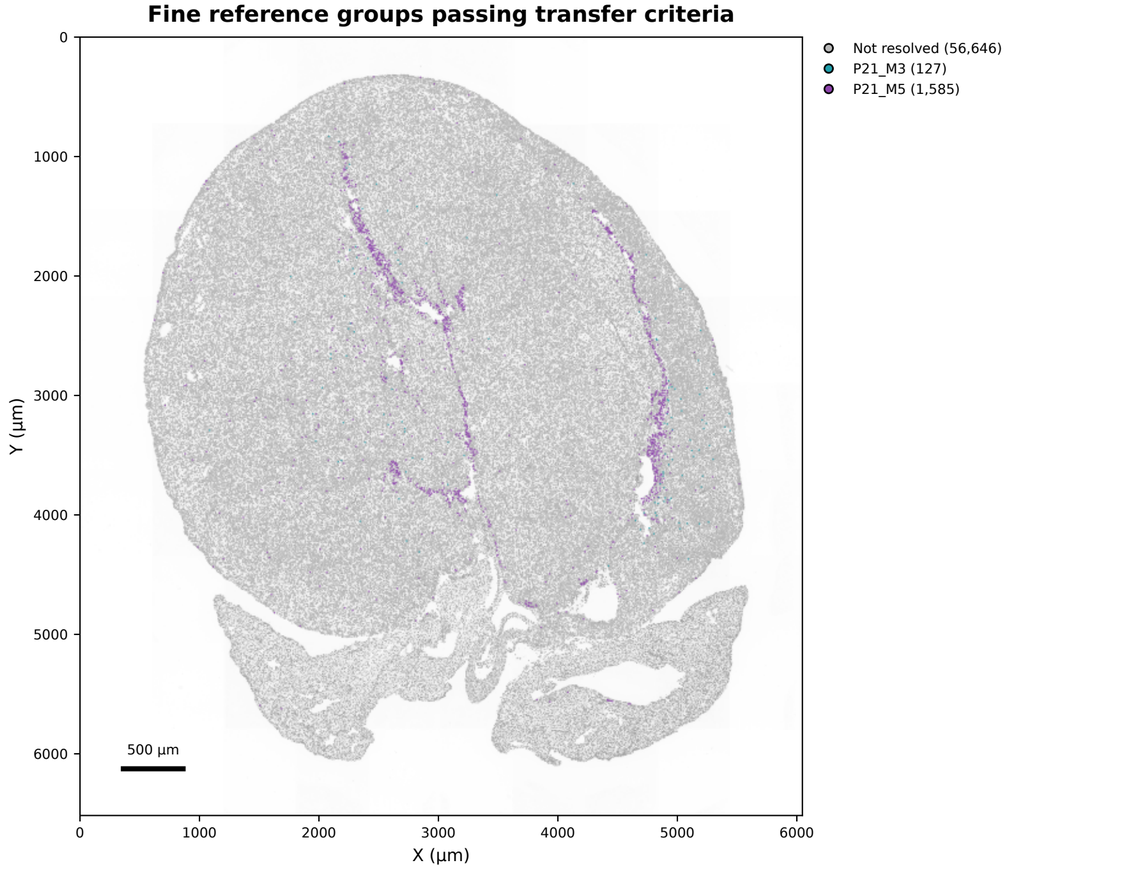
Measured-marker-supported lineage candidates



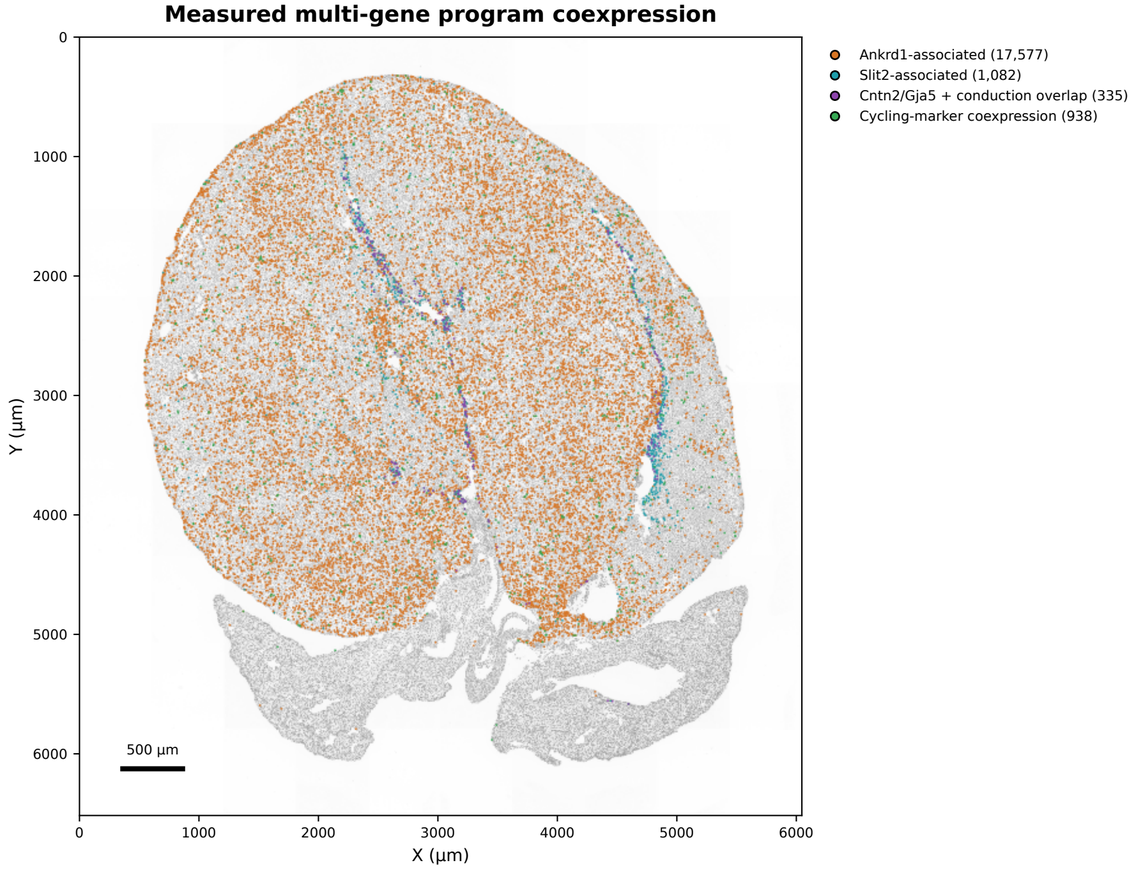
Ventricular program-family candidates



Fine reference groups passing transfer criteria

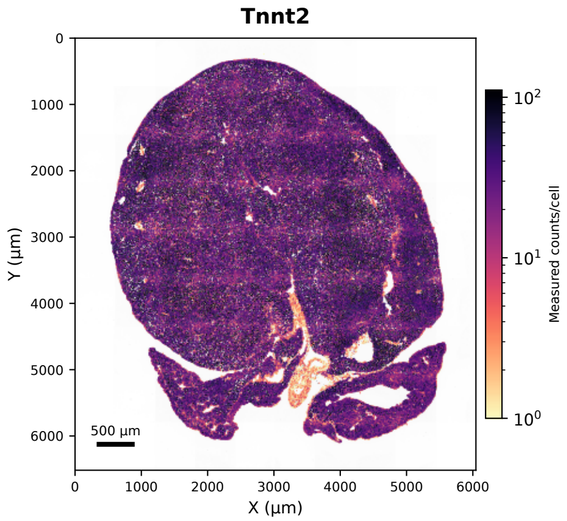


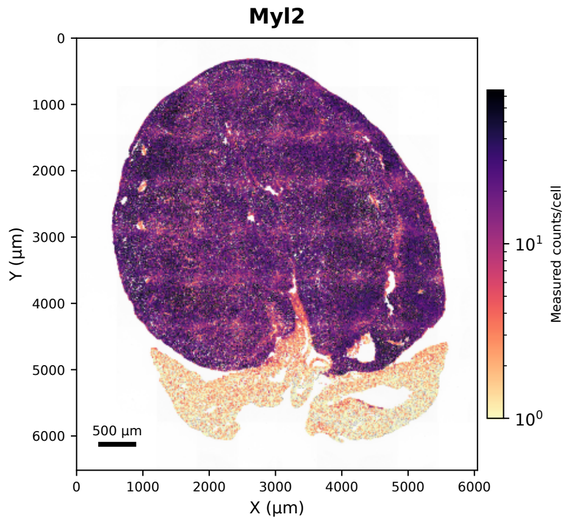
Measured multi-gene program coexpression

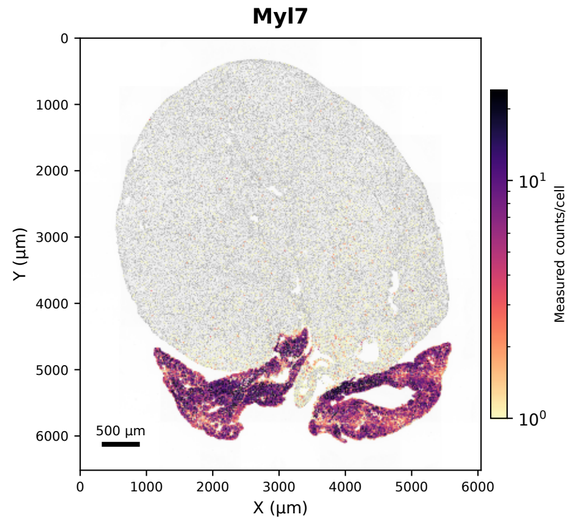


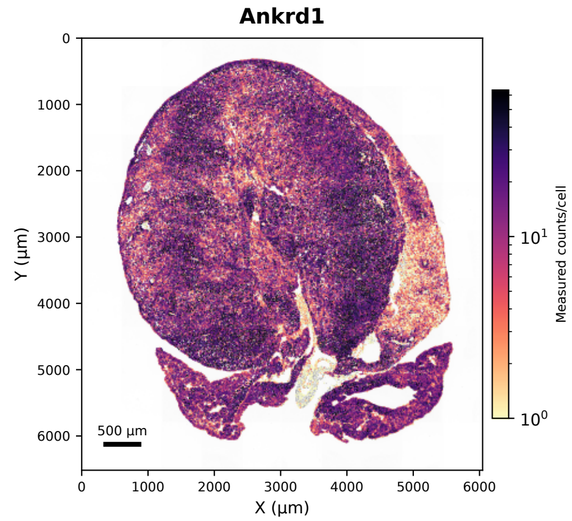
Gray image: acquired DAPI; colored points: original cell centroids. Candidate labels are reference-assisted, not anatomical or functional verification.
Fine assignments require panel/library/raw-reference agreement. Unresolved cells remain gray; measured program flags can overlap.
Kcnd2 and Irx5 are NOT assayed in this Xenium panel. No Kcnd2 expression, imputation or transmural coordinate is shown.

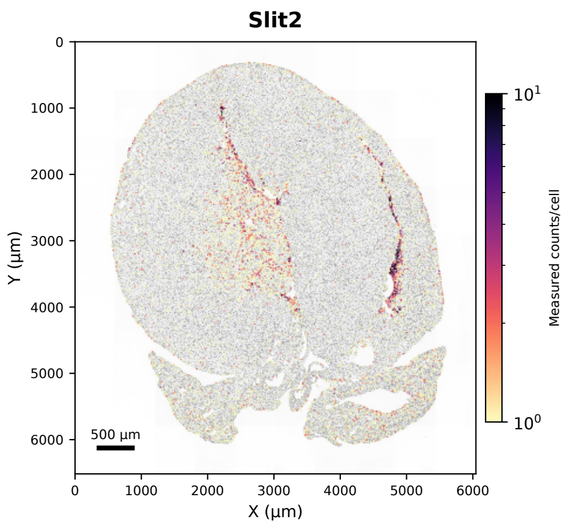
P21: directly measured Xenium RNA on DAPI

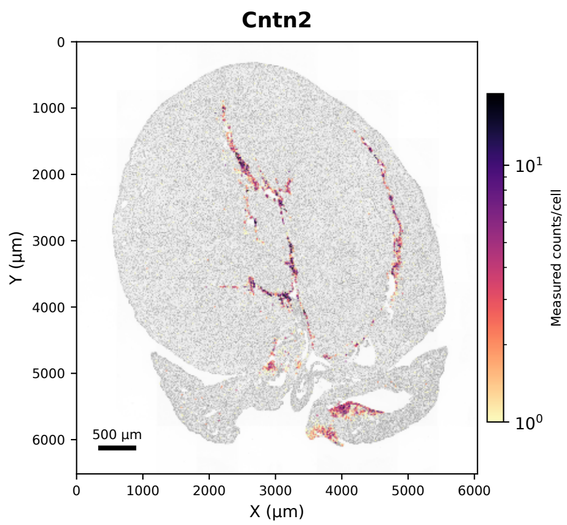


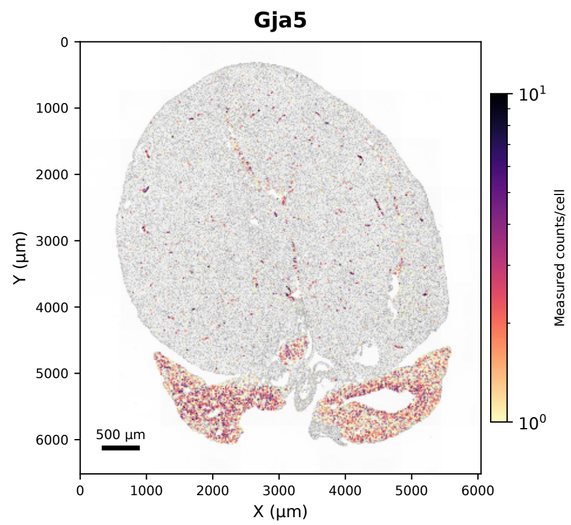


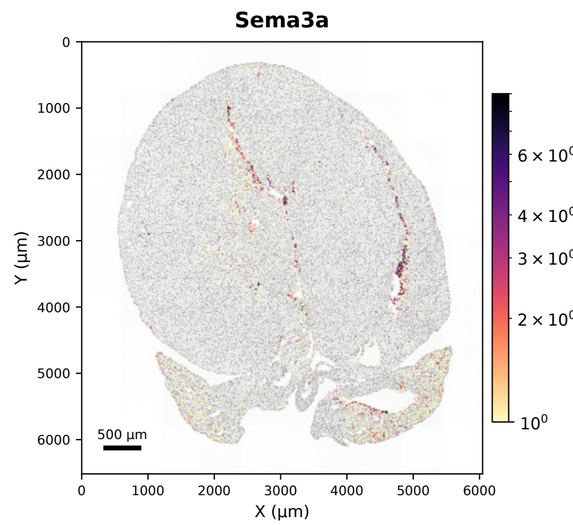


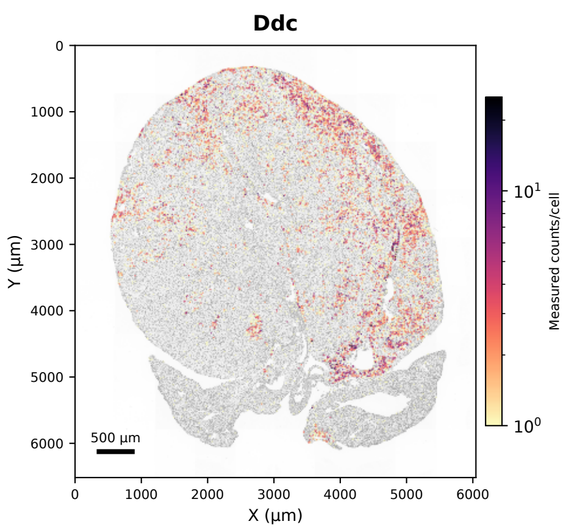


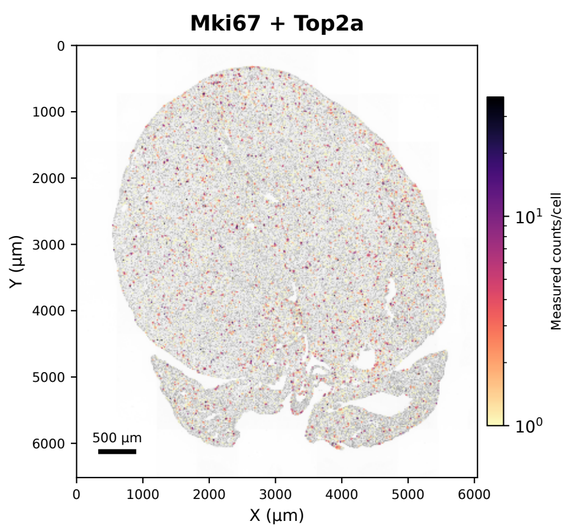


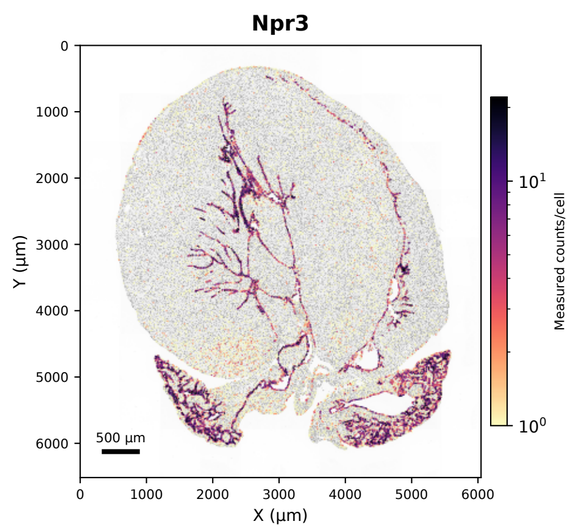


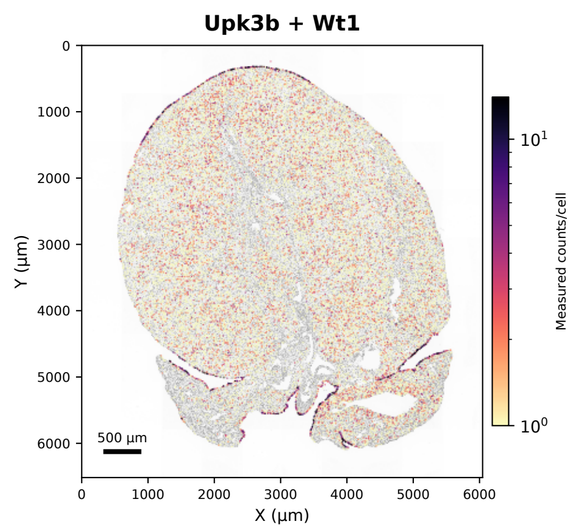






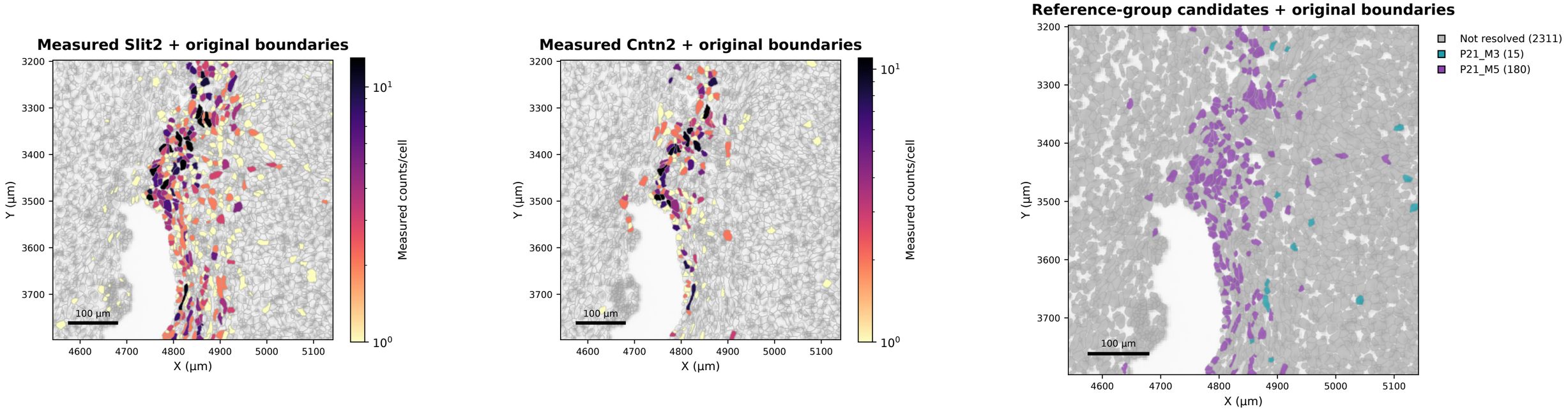






All QC-pass segmented cells shown; these genes are not all CM-specific (notably Gja5). Same gene uses the same scale across stages.
No count correction, smoothing or expression imputation. Marker counts depend on cell area, segmentation and detection efficiency.
Npr3 and Upk3b/Wt1 offer candidate anatomical landmarks; they are not traced wall boundaries. Kcnd2/Irx5 are absent from the panel.

P21: 600 × 600 µm illustrative region



ROI selected by density of reference-assisted conduction-program candidates; not an unbiased enrichment test.
Original boundaries and RNA counts retained; labels do not verify Purkinje identity. Kcnd2 is not measured.